THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 1

Chapter 3: **Metals and Non-metals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of the following pairs will give a displacement reaction? 1

(a) $Fe + CuSO_4$ solution **(b)** $FeSO_4$ solution and silver metal

(c) $MgCl_2$ solution and aluminium metal **(d)** NaCl solution and copper metal

2. The reactivity of sodium is explained by: 1

(a) its high melting point **(b)** its easy loss of an electron

(c) its hardness **(d)** its large mass

3. Metals like sodium and potassium are extracted from their ores by: 1

(a) heating alone **(b)** reduction with carbon

(c) electrolysis of molten chlorides **(d)** roasting

4. Which of the following methods is suitable for preventing an iron frying pan from rusting? 1

(a) applying paint **(b)** applying a coating of zinc

(c) applying grease **(d)** all of these

5. A metal that reacts with cold water to produce hydrogen but floats because of bubbles of hydrogen: 1

(a) magnesium **(b)** sodium **(c)** potassium **(d)** calcium

6. In the electrolytic refining of copper, the anode is made of: 1

(a) zinc **(b)** impure copper **(c)** graphite **(d)** pure copper

7. An element reacts with oxygen to give a compound with a high melting point, soluble in water. The element is likely to be: 1

(a) silicon **(b)** calcium **(c)** iron **(d)** carbon

8. Which of the following metals does not react with dilute HCl? 1

(a) zinc **(b)** copper **(c)** iron **(d)** magnesium

9. In the electrolytic refining of copper, the electrolyte is: 1

(a) NaCl solution **(b)** molten copper

(c) dilute HCl **(d)** acidified copper sulphate solution

10. Food cans are coated with tin and not with zinc because: 1

(a) zinc is cheaper than tin **(b)** zinc is less reactive than tin

(c) zinc is more reactive than tin **(d)** zinc has a higher melting point than tin

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Give an example of a metal which
(i) is a liquid at room temperature and
(ii) can be easily cut with a knife. — 2

12. What is aqua regia? Why is it important? — 2

13. Write the equation for the formation of sodium chloride showing electron transfer. — 2

14. Why is sodium kept immersed in kerosene oil? — 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A hammer, a battery, a bulb, wires and a switch are given. How will you distinguish between metals and non-metals using them? — 3

16. How is zinc obtained from its carbonate ore? Write equations. — 3

17. Differentiate between a metal and a non-metal in respect of
(i) conduction
(ii) oxides. — 3

OR

Write the equations for the reaction of — 3
(a) iron with steam
(b) calcium with water.

18. Why is mercury obtained from cinnabar simply by heating? Write the equations. — 3

OR

What is anodising? How does it protect aluminium? — 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain with equations the reactions of metals with oxygen, giving Na, Mg, Al and Cu as examples. State which oxide is amphoteric. — 5

OR

What is corrosion? Describe the conditions for rusting. State four ways to prevent it. — 5

20. Arrange Na, Mg, Al, Zn, Fe, Pb, Cu, Ag, Au in order of reactivity. Explain what happens when each class of metal reacts with water and acids. — 5

OR

What are the physical properties of metals? Give two exceptions. — 5

END OF QUESTION PAPER — Total: 40 Marks

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